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SUMMARY

Entry-level Data Scientist and Machine Learning enthusiast with a Civil Engineering background and professional project management experience. Completed hands-on training in Data Engineering, Machine Learning, and Data Science through IBM and DeepLearning.AI. Experienced in building ETL pipelines, performing data analysis, developing predictive models, and creating interactive dashboards using Python, SQL, Pandas, Scikit-learn, TensorFlow, PyTorch, and Dash. Skilled in developing end-to-end machine learning projects and deploying models through FastAPI and Docker. Passionate about solving real-world problems through data-driven solutions and continuously expanding technical expertise.

SKILLS

Programming

- Python
- SQL
- Bash
- JavaScript

Data Engineering

- ETL Pipelines
- REST APIs
- Web Scraping
- BeautifulSoup
- PostgreSQL
- MySQL
- SQLite
- Shell Scripting

MLOps / Deployment

- MLflow
- TensorBoard
- Docker
- Model Serving
- REST APIs
- Model Inference

Machine Learning

- Scikit-learn
- TensorFlow
- Keras
- XGBoost
- Regression
- Classification
- Clustering
- Feature Engineering
- Model Evaluation
- Cross Validation
- GridSearchCV
- Reinforcement Learning
- Recommender Systems
- Neural Network

Tools

- Git
- GitHub
- Jupyter Notebook
- VS Code

Deep Learning

- Pytorch
- CNN
- ResNet
- Image Classification
- Transfer Learning
- Computer Vision
- Image Preprocessing
- Fine-tuning

Web Development

- Javascript
- React
- Nodejs
- MongoDB
- ExpressJs
- REST API
- CSS
- Vite

Data Science & Visualization

- Pandas
- NumPy
- Matplotlib
- Seaborn
- GeoPandas
- Dash
- Folium
- Mathematics
- Statistics

Soft Skill

- Problem Solving
- Curiosity
- Team Player
- Flexible
- Good Communication
- Fast Learner

PROJECT

Bridge Crack Detection 🔗

PyTorch • ResNet18 • MLflow • FastAPI • Docker • React

- Developed an end-to-end computer vision application to classify bridge images as **crack** or **no crack** using a pretrained **ResNet18** model and transfer learning.
- Implemented image preprocessing, **data augmentation**, **class weighting**, early stopping, and **learning-rate scheduling** to improve model performance.
- Evaluated model performance using **accuracy**, **precision**, **recall**, **F1-score**, and **confusion matrix**.
- Integrated **TensorBoard** and **MLflow** for experiment tracking and monitoring of training metrics.
- Built a **FastAPI REST API** for image inference and containerized the application using **Docker**.
- Developed a **React frontend** that allows users to upload bridge images and receive the model's predicted class and confidence score.

<https://bridge-crack-detection.netlify.app/> 🔗

<https://github.com/nov-salvador/CNN-Learning-with-Bridge-Crack-Detection> 🔗

SpaceX Launch Success Prediction 🔗

Python • SQL • Scikit-learn • Dash • Folium

- Collected launch data through **REST APIs** and **web scraping**. Cleaned and transformed datasets using **Pandas**.
- Performed **SQL queries** for exploratory analysis. Built interactive visualizations with Matplotlib, Folium, and Dash.
- Developed and compared classification models including **Logistic Regression**, **Decision Tree**, **Random Forest**, **SVM**, and **XGBoost**. Optimized model performance using **GridSearchCV** and **cross-validation**.
- Created **comprehensive report** and **presentation** summarizing methodology and results.

https://github.com/nov-salvador/SpaceX_Landing 🔗

Weather Prediction Classification

Python • Pandas • Scikit-learn • Matplotlib

- Built a **classification model** to predict weather conditions based on past data. Performed **data preprocessing**, **feature engineering**, and **exploratory data analysis**. Trained and evaluated multiple **machine learning algorithms**.
- Assessed performance using metrics such as **Accuracy**, **Precision**, **Recall**, **F1-score**, and **Confusion Matrix**.

https://github.com/nov-salvador/Machine-Learning-with-Python---IBM

Titanic Survival Prediction

Python • Pandas • Scikit-learn

- Developed a **binary classification model** to predict passenger survival on the Titanic dataset. Handled missing values, **encoded categorical variables**, and **engineered new features**.
- Compared several machine learning models and evaluated performance using **cross-validation** and **classification metrics**.

https://github.com/nov-salvador/Machine-Learning-with-Python---IBM

Data Visualization

Python • Matplotlib • Dash • Folium • SQL

- Created interactive dashboards using **Dash**. Built geospatial visualizations with **Folium**. Produced statistical charts and exploratory analyses with **Matplotlib**. Communicated insights through effective **data visualization techniques**.

https://github.com/nov-salvador/Data-Visualization

Python ETL Pipeline

Python • Pandas • BeautifulSoup • SQLite

- Designed an ETL pipeline that extracts data from external sources, transforms it through **cleaning** and **mapping**, and **loads** it into **SQLite** and **CSV**. Automated preprocessing workflows using **Python** and **Pandas**.
- Demonstrated core data engineering concepts including **data extraction**, **transformation**, and **loading**.

https://github.com/nov-salvador/Python_Project_Data_Engineering_ETL

Web Development Portfolio

Full Stack Web Applications (MERN Stack) | KaMarites | Serbisio | AHAS | iLuto Mo

- Developed responsive full-stack web applications using **MongoDB**, **Express.js**, **React**, and **Node.js**.
- Implemented **REST APIs**, **authentication**, and **database integration**.

kamarites.netlify.app | serbisio.netlify.app | ilutomobaybe.netlify.app | ahas.netlify.app

EXPERIENCE

Project Engineer

EGB Construction Corp.

10/2024 – 11/2025

Iligan City, Isabela

- Analyzed project schedules and resource allocation to optimize workflow efficiency.
- Monitored project performance metrics to ensure timely completion and budget compliance.
- Coordinated multidisciplinary teams using data-driven planning and reporting.

Civil Engineer

J.F.E Techno Manila, Inc.

06/2023 – 09/2023

Pasig City, Metro Manila

- Evaluated structural designs using engineering principles and technical standards.
- Collaborated with multidisciplinary teams to improve project coordination.
- Conducted inspections and quality assurance to ensure compliance with specifications.

EDUCATION

Data Science Specialization

IBM

2026

Machine Learning Specialization

DeepLearning.AI & Stanford University

2026

Data Engineering Specialization

IBM

2026

Full Stack Web Development

Uplift Code Camp

2024

Bachelor of Science in Civil Engineering

University of Saint Louis

2022

Tuguegarao City,
Cagayan

CERTIFICATIONS

IBM Data Science Professional Certificate — IBM

Generative AI: Elevate Your Data Science Career — IBM

Machine Learning with Python — IBM

Data Visualization with Python — IBM

Data Analysis with Python — IBM

Data Science Methodology — IBM

Tools for Data Science — IBM

Machine Learning Specialization — DeepLearning.AI & Stanford University Online

Unsupervised Learning, Recommenders, Reinforcement Learning — DeepLearning.AI & Stanford University Online

Advance Learning Algorithm — DeepLearning.AI & Stanford University Online

Supervised Machine Learning: Regression and Classification — DeepLearning.AI & Stanford University Online

Relational Database Administration — IBM

Hands-on Introduction to Linux Commands and Shell Scripting — IBM

Databases and SQL for Data Science with Python — IBM.

Introduction to Relational Databases (RDBMS) — IBM

Python Project for Data Engineering — IBM

Python for Data Science, AI & Development — IBM

Registered Civil Engineer — 90.45% | PRC Philippines